PROJECT-1

Problem Title:-16-bit ALU Design with Basic Testbench and Flag Support

Project Description:-Design a 16-bit ALU capable of performing basic arithmetic and logical operations along with generation of essential status flags.

The goal is to evaluate:

Fundamental RTL design skills

Basic verification ability

Code organization and documentation

Problem Statement:-

The ALU should be capable of performing the following operations (minimum requirements):

Arithmetic Operations

- Addition (A + B)
- Subtraction (A − B)

Logical Operations

- AND
- OR
- XOR
- NOT (unary operation on A)

Shift / Rotate Operations

- Logical Left Shift
- Logical Right Shift

Status Flags (Mandatory)

The ALU must generate the following status flags:

- Z (Zero): Set when the result is zero
- C (Carry): Carry generated in addition or borrow in subtraction
- N (Negative): Most Significant Bit (MSB) of the result (sign bit)
- V (Overflow): Indicates signed overflow condition

1.Intoduction:-

An Arithmetic Logic Unit (ALU) is a fundamental component of digital systems and processors. It performs arithmetic and logical operations on binary data.

In this project, a 16-bit ALU is designed using Verilog HDL that supports multiple operations and generates status flags. The design is verified using a testbench and analyzed through simulation waveforms.

The goal of this project is to design a 16-bit Arithmetic Logic Unit (ALU) that can perform a set of arithmetic, logical, and shift operations on binary inputs.

The ALU must also generate status flags such as Zero, Carry, Negative, and Overflow, which are essential in processor operations.

In addition to design, the functionality must be verified using a testbench, and correctness must be validated using simulation waveforms.

2. Objective

- To design a 16-bit ALU using Verilog HDL
- To implement arithmetic, logical, and shift operations
- To generate status flags (Z, C, N, V)
- To verify the design using a testbench
- To analyze correctness using waveforms

3.Code :-

```
module null_gate(input [15:0]A,B, input [2:0]sel, output reg [15:0]Y ,output reg Z,output reg C, output reg N,output reg V);
always @(*) begin
Y=16'b0;
C=1'b0;
V=1'b0;
case(sel)
3'b000:
begin
{C,Y} = A+B;
V=((A[15]==B[15]) && (Y[15]!=A[15]));
end
3'b001:
begin
{C,Y} = A-B;
V=((A[15]!=B[15]) && (Y[15]!=A[15]));
end
3'b010: Y=A&B;
3'b011: Y=A|B;
3'b100: Y=A^B;
3'b101: Y=~A;
3'b110: Y=A<<1;
3'b111: Y=A>>1;
default: Y=16'b0;
endcase
if(Y==16'b0)
Z=1;
else
Z=0;
if(Y[15]==1)
N=1;
else
N=0;
end
endmodule
```

4.Testbench:-

```
module null_gate_tb;
reg [15:0]A,B;
reg [2:0]sel;
wire [15:0]Y;
wire Z;
wire C;
wire N;
wire V;
null_gate uut(.A(A),.B(B),.sel(sel),.Y(Y),.Z(Z),.C(C),.N(N),.V(V));
initial begin
A=16'd10;
B=15'd5;
sel=3'b000;
#10;
$display("ADD:Y=%d Z=%b C=%b N=%b V=%b",Y,Z,C,N,V);
sel=3'b001;
#10;
$display("SUB:Y=%d Z=%b C=%b N=%b V=%b",Y,Z,C,N,V);
sel=3'b010;
#10;
$display("AND:Y=%d",Y);
sel=3'b011;
#10;
$display("OR:Y=%d",Y);
sel=3'b100;
#10;
$display("XOR:Y=%d",Y);
sel=3'b101;
#10;
$display("NOT:Y=%d",Y);
sel=3'b110;
#10;
$display("LSHIFT:Y=%d",Y);
sel=3'b111;
#10;
$display("RSHIFT:Y=%d",Y);
$finish;
end
endmodule
```

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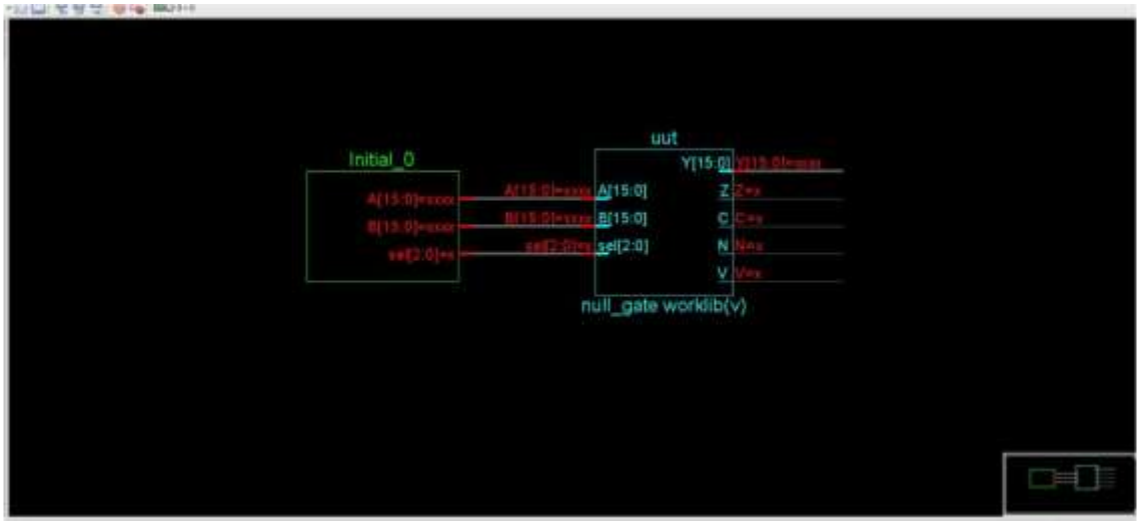
5. Features of the ALU

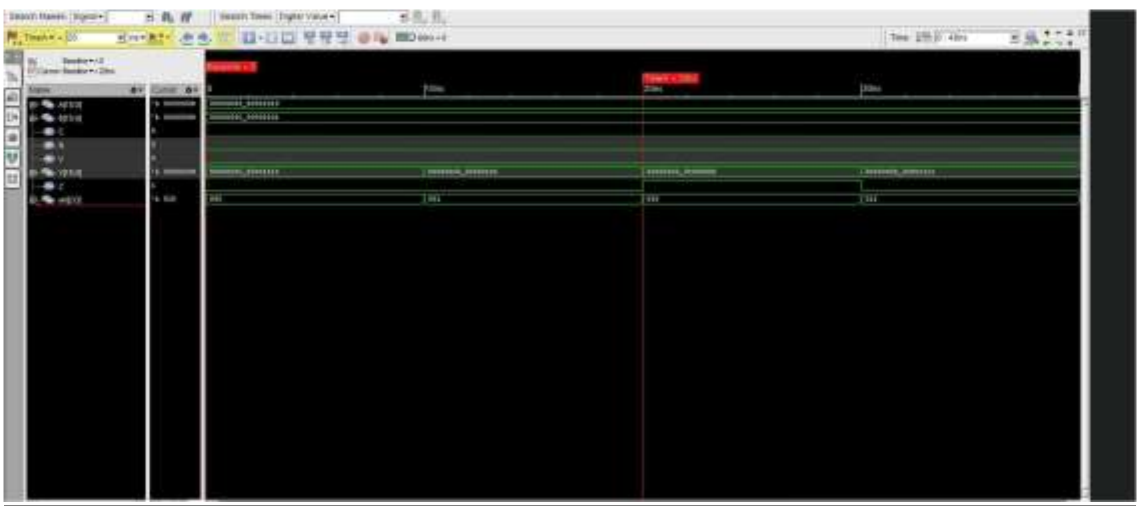
- 16-bit data processing
- Supports multiple operations using opcode
- Generates 4 status flags:
  - Zero (Z)
  - Carry (C)
  - Negative (N)
  - Overflow (V)

- Combinational design (no clock required)
- Easy to extend (parameterizable design possible)

6.Block Diagram:-



7.Waveform:-



8. Approach / Logic Used

The ALU is implemented using a combinational design approach in Verilog.

- A case statement is used to select operations based on the opcode
- Each operation (ADD, SUB, AND, etc.) is mapped to a unique opcode
- Arithmetic operations use built-in Verilog operators

- Flags are generated based on the result:
  - Zero (Z) → result equals zero
  - Negative (N) → MSB of result
  - Carry (C) → from addition/subtraction
  - Overflow (V) → detected using sign logic

The design ensures all outputs are updated instantly for any input change using always @(*).

8.Repository Link:-

https://github.com/nishitap2006-gif/rtl